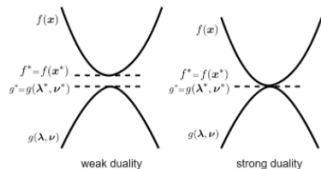
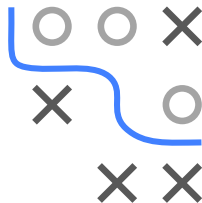


Optimization in Machine Learning

Nonlinear programs

Regularity Conditions



Learning goals

- KKT conditions
- Regularity conditions
- Examples

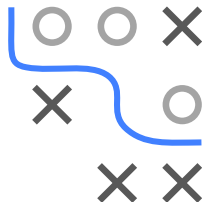
STATIONARY POINT OF THE LAGRANGIAN

- When we introduced the Lagrangian $\mathcal{L}$ from a geometrical perspective for the equality constraint problem, we realized that the geometrical conditions for the optimum coincided with finding a stationary point of $\mathcal{L}$:

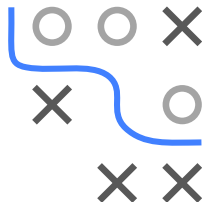
$$\begin{pmatrix} \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \beta) \\ \nabla_{\beta} \mathcal{L}(\mathbf{x}^*, \beta) \end{pmatrix} = \begin{pmatrix} \nabla f(\mathbf{x}^*) + \beta \nabla h(\mathbf{x}^*) \\ h(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- For the general Lagrangian, this leads to the following question:

Is $\nabla L(\mathbf{x}, \alpha, \beta) = 0$ a **necessary / sufficient condition for the optimum?**



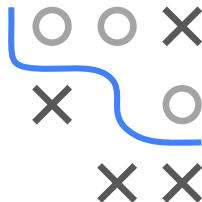
KKT CONDITIONS



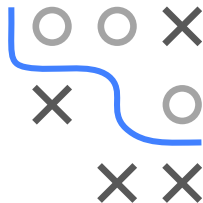
- To formulate necessary and sufficient conditions for optimality, we need the **Karush-Kuhn-Tucker conditions** (KKT conditions)
- A triple $(\mathbf{x}, \alpha, \beta)$ satisfies the KKT conditions if
 - $\nabla_{\mathbf{x}} L(\mathbf{x}, \alpha, \beta) = 0$ (stationarity)
 - $g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0$ for all i, j (primal feasibility)
 - $\alpha \geq 0$ (dual feasibility)
 - $\alpha_i g_i(\mathbf{x}) = 0$ for all i (complementary slackness)

KKT CONDITIONS

- **Necessary condition for optimality:** Let $\mathbf{x}^*$ be a local minimum. If certain regularity conditions are fulfilled, there are α^*, β^* such that $(\mathbf{x}^*, \alpha^*, \beta^*)$ fulfill the KKT conditions
- Under certain conditions, KKT conditions are also sufficient for optimality
- **Sufficient condition for optimality:** Given a **convex problem** (f convex, $\mathcal{S}$ convex) and $(\mathbf{x}^*, \alpha^*, \beta^*)$ satisfies the KKT conditions. Then $\mathbf{x}^*$ is a global solution to the problem



REGULARITY CONDITIONS



- Different regularity conditions (or constraint qualifications) ensure that the KKT conditions apply (ACQ, LICQ, MFCQ, Slater condition, ...)
- To use the above results, at least one regularity condition must be examined to prove that the function behaves “regular”
- We do not go further into these regularity conditions here

RIDGE REGRESSION

- The following two formulas are common for ridge regression:

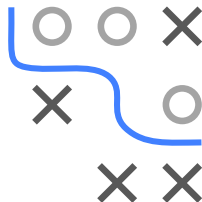
Formula 1:

$$\min_{\boldsymbol{\theta}} f_{\lambda}(\boldsymbol{\theta}) := \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2 \quad (1)$$

Formula 2:

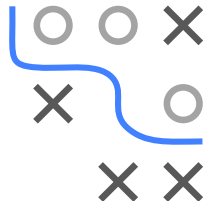
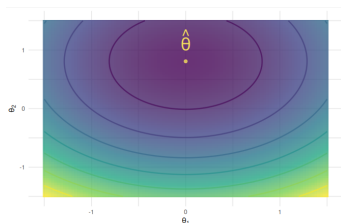
$$\begin{aligned} \min_{\boldsymbol{\theta}} \quad & \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \\ \text{s.t.} \quad & \|\boldsymbol{\theta}\|_2^2 - t \leq 0 \end{aligned} \quad (2)$$

- Why are these two formulas (for appropriate values t, λ) equivalent?

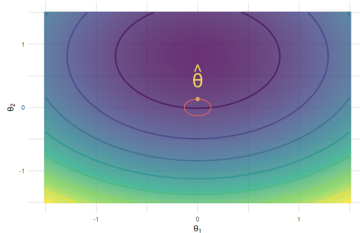
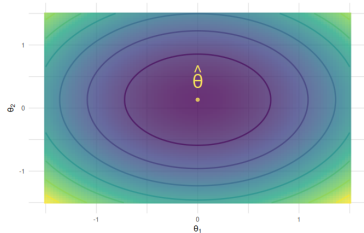


RIDGE REGRESSION – VISUALIZATION

- **Visualization:** see additional material



- Quadratic loss for the cars dataset without penalty



- Left: loss for ridge regression with penalty term
Right: loss for ridge regression with corresponding constraint

RIDGE REGRESSION – EQUIVALENCE

- Consider (1). If θ^* is our minimum, the necessary condition applies:

$$\nabla f_{\lambda}(\theta^*) = -2\mathbf{y}^T\mathbf{X} + 2(\theta^*)^T\mathbf{X}^T\mathbf{X} + 2\lambda(\theta^*)^T = 0$$

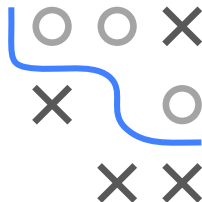
- We show that we can find a t so that θ^* is also solution for (2)
- We calculate the Lagrange function of (2):

$$L(\theta, \alpha) = \|\mathbf{y} - \mathbf{X}\theta\|_2^2 + \alpha(\|\theta\|_2^2 - t)$$

- The first KKT condition (stationarity) is:

$$\nabla_{\theta}L(\theta, \alpha) = -2\mathbf{y}^T\mathbf{X} + 2\theta^T\mathbf{X}^T\mathbf{X} + 2\alpha\theta^T = 0$$

- Since $\nabla f_{\lambda}(\theta^*) = 0$, this is fulfilled if we set $\theta = \theta^*$ and $\alpha = \lambda$



RIDGE REGRESSION – EQUIVALENCE

- However, complementary slackness must still apply for the KKT conditions:

$$\alpha(\|\boldsymbol{\theta}\|_2^2 - t) = 0$$

- This is the case if we choose $t = \|\boldsymbol{\theta}^*\|^2$
- Vice versa it can be shown that a solution of (2) is a solution of (1) if we set $\lambda = \alpha$

